

DATA MANAGEMENT PLAN

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INITIAL PLAN

The original DMP proposed three research questions, but the final project focused primarily on RQ1: *How can community-curated datasets be structured using database design and FAIR metadata standards?* To address this question, the project developed:

- A 7-table relational database normalized to **3NF**
- A 3-level genre taxonomy (**9 super-genres -> 78 genres -> 41 sub-genres**) to support the ERD
- A **SKOS crosswalk** between Steam folksonomy tags and MobyGames controlled vocabulary

The scope expanded to include a systematic comparison of community tagging and library classification systems, an addition prompted by class feedback.

WHAT CHANGED / Dataset

Initial DMP	Final Implementation	Reason
Seyed's Publisher List	Gamalytics publisher data processed by Seyed (205 publishers)	Correct attribution: the dataset was provided by Gamalytics
Steam Trends 2023 (Barabanov & Kobelev)	Top 1000 Steam Games 2024–2026 (Waddah Ali, Kaggle, CC BY-NC-SA 4.0)	The original dataset contained systematic errors in release-date formatting, rendering temporal analysis unreliable.
No crosswalk planned	MobyGames genre categories + SteamDB tag filter (manual crosswalk)	Taxonomy development revealed the need to compare Steam folksonomy with a controlled vocabulary.

WHAT CHANGED / Tools

Initial DMP	Final Implementation	Reason
R (statistical analysis)	Python only (pandas, matplotlib, networkx)	Python handled the required analysis, including network graphs, co-occurrence counting, and tag classification.
Tableau (visualization)	Observable (sunburst)	Visualization needs have shifted with the stronger focus on taxonomy and folksonomy analysis.
Lucidchart (diagrams)	Still apply Lucidchart as a reference	Lucidchart supported the structural planning of the taxonomy and 3NF ERD.

Zenodo (repository)	OSF, Open Science Framework	OSF provided a DOI assignment and a simpler public upload workflow.
OCLC WorldCat API	Not used	Time constraints and the metadata's limited relevance led to its exclusion.

DELIVERY MATRIX

Component	Fair	Good	Excellent
Documentation	README.md with attribution and basic instructions	README.md PLUS data dictionary	Data dictionary PLUS complete documentation package ready for sharing
Entity-Relationship Diagrams	Conceptual ERD	Conceptual PLUS Logical ERD	All three ERDs (Conceptual, Logical, Physical) with SQL scripts
Taxonomy & Ontology	Controlled vocabulary list	List PLUS basic taxonomy diagram	Complete taxonomy and ontology diagram with relationships
Analytics & Visualization	Descriptive statistics with findings	Statistics + charts	Statistics + charts + interactive Tableau dashboard

Green blocks were delivered, yellow ones were adjusted for the folksonomy-taxonomy analysis, and the red block was discarded. The interactive Tableau dashboard was replaced. The publisher—genre and popularity—quality analyses were not completed, but the database structure preserves these as future steps.

REFLECTION

The project delivered more depth on RQ1 than originally planned, and less breadth overall. The decision to focus on the taxonomy and folksonomy comparison, which provides a more coherent and defensible contribution between the community and the structured dataset. The core FAIR question was answered with a complete, reproducible, documented pipeline. The unfinished analyses (publisher-genre, popularity-quality) are set up in the database and represent clear next steps rather than abandoned work.